Designing X-ray Optics for Pulsar-informed Navigation

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INTRODUCTION

Millisecond pulsars exhibit predictable long-term stability, comparable to atomic clocks, and can serve as natural beacons for autonomous position and timing systems.[1,2,3] We present the results from phase-resolved spectral analysis of NICER measurements for a set of millisecond pulsars that are prime candidates for Pulsar-informed Navigation (PIN). This analysis shows the energy-dependent signal to noise ratio of the pulsed emission. Customized multi-layer coatings can be designed to reflect photons in narrow energy bands across a wider range of grazing incidence angles, enabling shorter focal lengths for more compact telescopes. We discuss the design space and show initial results from instrument design tools for a PIN instrument.

Pulsars

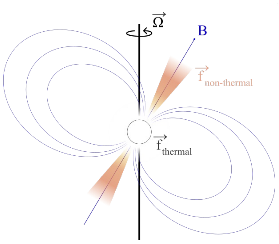

Fig. 1: Illustration of a pulsar and its magnetic filed B, offset from the axis of rotation Ω.

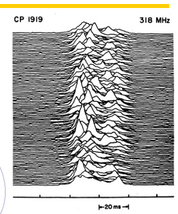

Fig. 2: Radio recording of Pulsar CP 1919 (PSR B1919+21) [4] showing variation of individual pulse profiles, but steady phase.

The potential for a compact X-ray instrument was identified in 1981 before the first millisecond pulsars (MSPs) were detected in 1982.[5,6] That concept was realized in 2017 with the design of a 0.19 m^2 optic with < 300 ns timing resolution with NICER which accomplished 5 km accuracy due to the stability of MSPs.[7,8] However, an order of magnitude reduction in the instrument size and two orders of magnitude improvement in position accuracy remains a viability gap compared to GPS, radio, and laser ranging.[9]

Pulsar Timing for Navigation

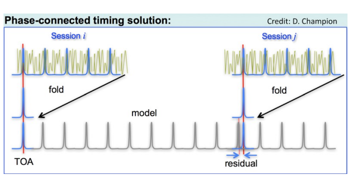

Fig. 3: Diagram of two observations of a pulsar, showing folding data into one phase. Observation Session *i* establishes a timing model to predict forward. Session *j* later compares with the model prediction and measures a residual offset.

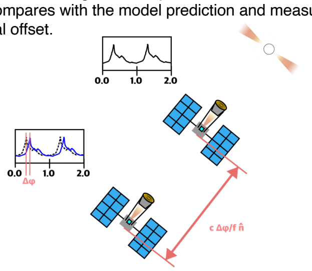

Fig. 4: Illustration of pulsar-informed navigation. Two observations of a distant pulsar are made at different locations, yielding characteristic pulse profiles (inset histograms of counts vs phase, two cycles shown). The spatial separation between the observations introduces a measurable phase shift, Δϕ, between the peaks of the measured pulses. The physical distance along the direction to the pulsar, n̂, is directly related to this phase shift by the relation cΔϕ/f , where c is the speed of light and f is the pulsar rotation frequency.

PHASE-RESOLVED SPECTROSCOPY OF MILLISECOND PULSARS

Each millisecond pulsar has a unique spectra and energy at which the pulsed emission from the pulsars is most visible.

Data is available from NICER observations of many X-ray MSPs. Software tools also already exist including HEASoft and its NICERDAS package of mission-specific functions, and PINT, a pulsar timing tool that can reference parameter files created from radio observations by NANOGrav and recover pulse profiles from NICER data of sources. [10]

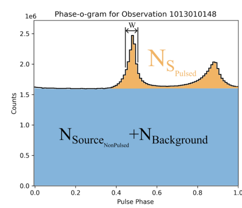

Fig. 5: A histogram of counts versus pulse phase for Observation ID 1013010148 of the Crab Pulsar, PSR B0531+21. The non-pulsed source emission and background emission are indistinguishable and combine to fill the static blue region of the plot. The Pulsed source emission forms the dynamic orange region at the top of the plot. The pulse width, W, is annotated. Note that counts on the Y-axis are scaled by one million.

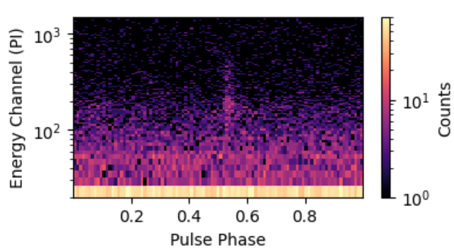

Fig. 6: A histogram of counts by phase, separating counts out by energy channel. This figure was created from a NICER telescope observation of the pulsar B1937+21. Energy Channels are 10 eV wide, so 10^2 represents 1 keV, and 10^3 represents 10 keV. A pulse peak is visible near Pulse Phase 0.55 and is prominent between 1 keV and 5 keV.

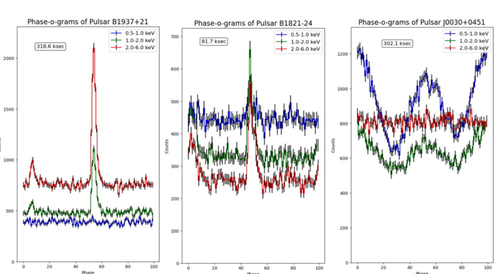

Fig. 7a,b,c: Each of the pulsars exhibit maximum counts of pulsed emission in a different energy band. PSR B1937+21 and B1821-24 have narrow peaks, while PSR J0030+0451 has a broad peak that is not strongly detected above 1.0 keV and intermingles with soft X-ray background and non-pulsed emission.

INSTRUMENT DESIGN

Single and multilayer coatings are modeled on a range of instrument parameters. We model X-ray optics using open source code to define a geometry (Nested Optics Designer) and to estimate the reflectivity of an array of mirrors using (pyreflectivity) and LBNL CRXO webtools [11].

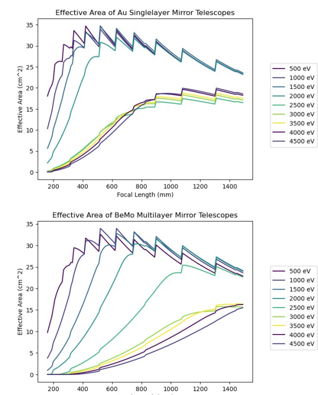

Fig. 8: Effective area of Wolter-I mirrors with parabolic and hyperbolic stages, maximally nested shells, and parameters in the following table. Boosts in area are a result of nesting an additional shell given that focal length. Energy bands show how lower energies are maximally reflected by larger geometric area found with higher incidence angles and sufficient reflectivity.

Model Parameter	Value
Primary Radius (Reference)	25 mm
Mirror Thickness	1.1 mm
Mirror Length	100 mm
Maximum Radius	50 mm
Minimum Radius	15 mm
Minimum Spacing	0.0 mm

Multilayer Parameter	Value
Material	Be/Mo
Period	0.438 nm
Number of Periods	70
Ratio (bottom layer thickness/period)	0.3

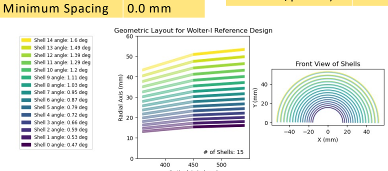

Fig. 9: Cross sections of a Wolter-I telescope design with parameters described in the Model Parameter table for 450 mm Focal Length. Includes fifteen complete shells, each with parabolic and hyperbolic segments.

CONCLUSIONS

Analysis of three MSPs (12 to 20 observations each, spanning seven years, with 80 to 300+ ksec of exposure per pulsar) show that these sources each exhibit different emission characteristics. PSR B1937+21 and B1821-24 are promising but not identical candidates for PIN, and their emission characteristics will be considered as design parameters for a PIN instrument. PSR J0030+0451 has a wider pulse, strong secondary pulse, and low energy making it a non-ideal reference source for PIN, but indicating unique physical properties.

A multilayer coating optimized for this energy and these incidence angles could further improve the reflectivity specifically of pulsars for navigation.

REFERENCES

1. Hewish et al 1968
2. Sagan et al 1972
3. Allan 1987
4. H. Craft 1970
5. Chester & Butman 1981
6. Backer et al 1982
7. Gendreau et al 2016,
8. Mitchell et al 2018
9. Sheikh et al 2006
10. Agazie et al 2023
11. Henke et al 1993